STAT 5200, Notes 6[1]

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Readings. Section 4.3 in W. The section deals with omitted explanatory variables in the context of OLS estimation. Often the reason a variable is omitted from a regression calculation is that it has not been or cannot be observed. This omission can lead to bias in the estimation of partial slopes for the other variables. The bias can usually be lessened if a proxy for the omitted variable is available. Section 4.3 also addresses interaction between an omitted explanatory variable and an observed explanatory variable.

OLS and omitted variables. Let's assume the model describing the relation between the response and explanatory variables is

$$\begin{aligned} y &= E[y|x_1, x_2, ..., x_K, q] + v \\ &= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_K x_K + \gamma q + v, \end{aligned} \tag{1}$$

where the x_k's are observed and q is unobserved. Note the conditional expectation gives the *true model*, and the conditional expectation is also the linear projection (Property LP.1 in W). Also note that, by properties of a linear projection, v has mean zero and is uncorrelated with the x_k's and also uncorrelated with q (Property LP.2 in W; this also follows from the conditional expectation and Property CE.6 in W). Suppose we write

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_K x_K + u = \mathbf{x}\beta + u, \tag{2}$$

where $\mathbf{x} = (1, x_1, x_2, ..., x_K)$, $\beta = (\beta_0, \beta_1, \beta_2, ..., \beta_K)$' and

$$u = \gamma q + v. \tag{3}$$

That is, we add γq and v and view (3) as the error term. Then we fit the regression of y vs. $1, x_1, x_2, ..., x_K$, as suggested by (2). This, however, poses a potential problem. Typically the unobserved variable q will be correlated with the x_k's, and thus the error term in (2) is correlated with the explanatory variables. That is, there is endogeneity, and the estimates of all the β_j's will be biased. Let's characterize the bias.

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I'll do all these calculations in the population. Hence the argument given relates to asymptotic theory – it addresses asymptotic bias. If there is bias asymptotically, we also expect to experience bias in finite samples. Suppose q is correlated with the x_k's and we estimate (2) by OLS. We form the linear projection of y on $\mathbf{x} = (1, x_1, x_2, ..., x_K)$. This has the form

$$L(y|\mathbf{x}) = \mathbf{x}\eta, \quad \text{where } E[\mathbf{x}'\mathbf{x}]\eta = E[\mathbf{x}'y]. \tag{4}$$

Then

$$\eta = \left[E[\mathbf{x}'\mathbf{x}]\right]^{-1} E[\mathbf{x}'y] = \left[E[\mathbf{x}'\mathbf{x}]\right]^{-1} E[\mathbf{x}'(\mathbf{x}\beta + u)] = \beta + \left[E[\mathbf{x}'\mathbf{x}]\right]^{-1} E[\mathbf{x}'u], \tag{5}$$

and there is bias unless u is uncorrelated with *all* of the x_k's.

As an aside, let's consider what property LP.2 in W says in the present context. We need to fully understand this property for the discussion below. Property LP.2 states that the residual from a projection is always uncorrelated with the explanatory variables. Let's prove this in the notation of the present context. The residual from the linear projection we are considering is

$$w = y - L(y|\mathbf{x}) = y - \mathbf{x}\eta, \tag{6}$$

and then

$$E[\mathbf{x}'w] = E[\mathbf{x}'(y - \mathbf{x}\eta)] = E[\mathbf{x}'y] - E[\mathbf{x}'\mathbf{x}]\eta = \mathbf{0},$$

the last equality following immediately from (4). Because the first component of $\mathbf{x}$ is 1, it is also true that $E[w] = 0$. Note that w defined in (6) is *not the same* as u defined in (2) and (3) and η is not the same as β in (2).

Let's use scalar notation to clarify the form of the bias when we use OLS with (2) to estimate $\beta = (\beta_0, \beta_1, \beta_2, ..., \beta_K)'$ and the true model is (1). Form the linear projection of the missing variable q onto the observed explanatory variables,

$$q = \delta_0 + \delta_1 x_1 + \cdots + \delta_K x_K + r, \tag{7}$$

where $E[r] = 0$ and $Cov(x_k, r) = 0$, $k = 1, ..., K$. Now substitute (7) into (1), yielding

$$y = (\beta_0 + \gamma\delta_0) + (\beta_1 + \gamma\delta_1)x_1 + (\beta_2 + \gamma\delta_2)x_2 + \cdots + (\beta_K + \gamma\delta_K)x_K + \gamma r + v. \tag{8}$$

The error term in (8) is $v + \gamma r$; it has zero mean and is uncorrelated with each of the x_k's. Equation (8) shows that OLS with (2) is inconsistent for estimation of all the partial slopes attached to the x_k's, and it also shows what the biases are. In a situation where context gives some information about the algebraic signs and magnitudes of γ and of the δ_k's, we have some understanding of the directions and magnitudes of the biases. The regression (7) is considered so q can be decomposed into two pieces, that part which is correlated with the x's and that part which is uncorrelated with the x's. This allows us to see clearly the biases in OLS regression with (2).

W's Example 4.2 treats a wage equation with education and experience as observed explanatory variables, and ability as an unobserved explanatory variable.

Use of a proxy variable. A proxy for the unobserved explanatory variable q is sometimes available. Wooldridge specifies two requirements for z which make it a *good* proxy for q. One is

$$E[y|\mathbf{x}, q, z] = E[y|\mathbf{x}, q]. \tag{9}$$

This states that, once $\mathbf{x}$ and q have been accounted for, z provides no additional predictive value. The second requirement is

$$L[q|1, x_1, ..., x_K, z] = L[q|1, z]. \tag{10}$$

This will hold if z is sufficiently highly correlated with q so that the x_j's cannot capture any additional partial correlation. In applications where we have selected a proxy, (10) will seldom be satisfied, but with a decent proxy, the partial slopes attached to the x's will be relatively small in magnitude.

In W's Example 4.3, IQ score is a proxy for ability. It follows from (10) that we can write (if z is a good proxy),

$$q = \theta_0 + \theta_1 z + r, \tag{11}$$

and $E[r] = 0$ and $Cov(z, r) = 0$. If z is a good proxy, θ_1 will not be 0, and usually we think of it as being positive. Assumption (10) also ensures that $Cov(x_j, r) = 0$, $j = 1, ..., K$.

From (1) and (11) we can write

$$y = (\beta_0 + \gamma\theta_0) + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_K x_K + \gamma\theta_1 z + \gamma r + v. \tag{12}$$

The error term in (12) is $\gamma r + v$, and it is uncorrelated with all the x_k's and with z by property LP.2 and assumptions (9) and (10). Thus, a good proxy permits consistent estimation of the partial slopes attached to the x_k's in (1).

What if the proxy is imperfect? Then instead of (11) we have

$$q = \theta_0 + \rho_1 x_1 + \rho_2 x_2 + \cdots + \rho_K x_K + \theta_1 z + r, \tag{13}$$

and OLS gives inconsistent estimation of the partial slopes attached to the x_k's in (1). However, if the ρ_j are relatively small, there will usually be some benefit from the use of z as a proxy. Furthermore, the use of a proxy can lessen asymptotic variances of the parameter estimates as well as reduce bias. See the discussion on page 69 of W.

Appendix. Use of proxy when there are interactions involving unobserved variables requires some caution. Recall that the first requirement for a good proxy is

$$E[y|\mathbf{x}, q, z] = E[y|\mathbf{x}, q] \tag{14}$$

(q is the unobserved variable and z is its proxy). As noted above, this states that, once $\mathbf{x}$ and q have been accounted for, z provides no additional predictive value. The second requirement for a good proxy was stated above as a linear projection result,

$$L(q|1, \mathbf{x}_1, ..., x_K, z) = L(q|1, z) \tag{15}$$

where $\mathbf{x} = (1, x_1, ..., x_K)$. This will hold if z is sufficiently highly correlated with q so that the x_j's cannot capture any additional partial correlation. From (15) we can write

$$q = \theta_0 + \theta_1 z + r, \tag{16}$$

and $E[r] = 0$ and $Cov(z, r) = 0$.

Suppose the model we want to fit is a linear projection plus error,

$$\begin{aligned} y &= L(y|\mathbf{x}, q) + v \\ &= \beta_0 + \beta_1 x_1 + \cdots + \beta_{K-1} x_{K-1} + \beta_K x_K + \gamma_1 q + \gamma_2 x_K q + v. \end{aligned} \tag{17}$$

Here v is uncorrelated with the elements of $\mathbf{x}$ and with q, and the linear projection includes an interaction term involving the unobserved variable.

First, let's note that, if we assume

$$E[q|\mathbf{x}] = 0, \tag{18}$$

then both q and $x_K q$ are uncorrelated with all of the elements of $\mathbf{x}$, and we estimate all of the β's consistently by fitting the regression

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_{K-1} x_{K-1} + \beta_K x_K + u, \tag{19}$$

where

$$u = \gamma_1 q + \gamma_2 x_K q + v \tag{20}$$

is the error term. Of course the presence of $\gamma_1 q + \gamma_2 x_K q$ in the error term will inflate standard errors.

Suppose we substitute (16) into (17). This yields

$$\begin{aligned} y &= \beta_0 + \beta_1 x_1 + \cdots + \beta_{K-1} x_{K-1} + \beta_K x_K + \gamma_1(\theta_0 + \theta_1 z + r) + \gamma_2 x_K(\theta_0 + \theta_1 z + r) + v \\ &= \beta_0 + \gamma_1\theta_0 + \beta_1 x_1 + \cdots + \beta_{K-1} x_{K-1} + (\beta_K + \gamma_2\theta_0) x_K + \gamma_1\theta_1 z + \gamma_2\theta_1 x_K z + \gamma_1 r + \gamma_2 x_K r + v. \end{aligned} \tag{21}$$

There are two problems with this formulation. First, the presence of the parameter θ_0 results in the partial slope attached to x_K being $\beta_K + \gamma_2\theta_0$, instead of β_K. But, in addition, the error term contains $\gamma_2 x_K r$, with the result that (potentially) all of the explanatory variables are correlated with the error term. The formulation would still not be viable if θ_0 were zero.

To repair this situation, we need to switch from a linear projection formulation to one based on conditional expectation, and we need to assume both q and z have mean zero. The latter assumption will be achieved if we demean the proxy z. We now assume (14) and, in place of (15),

$$E[q|\mathbf{x}, z] = E[q|z] = \theta_1 z. \tag{15'}$$

Then the analog of (17) is

$$\begin{aligned} y &= E[y|\mathbf{x}, q] + v \\ &= \beta_0 + \beta_1 x_1 + \cdots + \beta_{K-1} x_{K-1} + \beta_K x_K + \gamma_1 q + \gamma_2 x_K q + v. \end{aligned} \tag{17'}$$

(Note that the above states that the conditional expectation is a linear function).

By (14), (15'), and iterated expectations,

$$\begin{aligned} E[y|\mathbf{x}, z] &= E\left[E[y|\mathbf{x}, q, z] \mid \mathbf{x}, z\right] \quad \text{, by the law of iterative expectation} \\ &= E\left[E[y|\mathbf{x}, q] \mid \mathbf{x}, z\right] \quad \text{, by the first requirement of a good proxy} \\ &= \beta_0 + \beta_1 x_1 + \cdots + \beta_{K-1} x_{K-1} + \beta_K x_K + \gamma_1 E[q|\mathbf{x}, z] + \gamma_2 x_K E[q|\mathbf{x}, z] \quad \text{, by (17')} \\ &= \beta_0 + \beta_1 x_1 + \cdots + \beta_{K-1} x_{K-1} + \beta_K x_K + \gamma_1 \theta_1 z + \gamma_2 \theta_1 x_K z, \quad \text{, by (15')} \end{aligned} \tag{22}$$

and all the β parameters can be estimated consistently in this formulation.